Section 5 Solution

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3 Exercise

Problem 1 (Problem 2 from PS1, Spring 2014)

Let a and b be two real non null parameters. Consider the following maximization problem: $max_{x,y} f(x, y; a, b) = ax^2 - x + by^2 - y$

1. Write down the first order conditions for this problem with respect to x and y (notice that a and b are parameters).
2. Solve explicitly for the x^* and the y^* that satisfy the first order conditions.
3. Now the second order conditions at (x^*, y^*). Give sufficient conditions on a and b for the stationary point that you found in point 2 to be a maximum?
4. Under what conditions on a and b is the function f concave in x and y? When is it convex in x and y?

Solution:

1. The first order condition with respect to x is $2ax - 1 = 0$. The first order condition with respect to y is $2by - 1 = 0$.
2. Solving the first order conditions yields: $x^* = \frac{1}{2a}$ and $y^* = \frac{1}{2b}$
3. We first compute the Hessian matrix:

   $$H_f = \begin{bmatrix} 2a & 0 \\ 0 & 2b \end{bmatrix}$$

   The second order conditions for a local maximum are that $2a < 0$ and $4ab > 0$, where $4ab$ is the determinant of the Hessian.

   These inequalities imply that the function assumes a local maximum at (x^*, y^*) whenever $a < 0$ and $b < 0$.

   Furthermore, since this is the only point that satisfies the first order conditions, it follows that it is the only local maximum.

   Lastly, the fact that $a, b < 0$ implies that as either x or y tend to infinity, the function f becomes increasingly negative.

   It follows that whenever $a, b < 0$, the function reaches a global maximum at (x^*, y^*).
4. Recall the Hessian matrix from above.

   The conditions for concavity are $a < 0$ and $4ab > 0$.

   The conditions for convexity are $a > 0$ and $4ab > 0$.

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Problem 2

Consider the function f defined on $[-1,1] \times [-1,1]$ by: $f(x,y) = x^2 - y^2$.

1. Provide the maximal points of f and the points (x,y) at which the maximum is reach (no calculus needed).

2. Let's now try to use our recipe:

   (a) Write down the first order conditions for this problem with respect to x and y and solve for the x^* and the y^* that satisfy the first order conditions.

   (b) Compute the hessian of f at the stationary points you found. Is the hessian definite positive? Is it negative? What can you infer?

Solution:

1. For any $(x,y) \in [0,1]^2$: $x^2 \leq 1$ and $y^2 \geq 0$. Therefore, for any $(x,y) \in [0,1]^2$: $f(x,y) \geq 1$.

   Let's solve the equation $f(x,y) = 1$ for x and y:

   $$f(x,y) = 1 \Rightarrow y^2 = x^2 - 1 \tag{1}$$

   But we know, on one side $y^2 \geq 0$ and on the other side: $x^2 - 1 \leq 0$, therefore $y^2 = 0$. Hence, for any solution to our equation: $y = 0$, implying that $x^2 = 1$ (or equivalently $x = 1$ or $x = -1$).

   One can conversely check that f reaches 1 at $(1,0)$ and $(-1,0)$.

2. (a) The first order condition for x is: $2x = 0$, equivalent to $x = 0$.
   The first order condition for y is: $-2y = 0$, or $y = 0$.

   (b) The hessian of f at the stationary point:

   $$H_f = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}$$

   H_f is neither positive nor negative. There is nothing we can infer about the stationary point (neither that it is a local optimum nor that it is not a local maximum. However, given the answer to question (1), and given that $f(0,0) = 0$, we know the stationary point does not achieve a global maximum).

Problem 3

Suppose you are taking only two classes: literature and economics. It is finals week and you know you can study productively for T hours a day. You must choose how to split these hours between studying for literature (which involves a lot of reading) and studying economics. Your goal is to maximize your GPA. An exogenous variable, the amount of free coffee you get, helps you read more efficiently– but it does not affect your studying of economics. Assume that you know exactly how your daily studying of each subject will determine your GPA this semester:

$$GPA = \frac{2}{3}\sqrt{C}\sqrt{h_L} + \frac{2}{3}\sqrt{h_E}$$

where h_L = hours studying literature, h_E = hours studying economics, and C = cups of coffee.

1. What would be the Lagrangian for this maximization problem? Find the potential optimum and check the second order condition.

2. How will the hours spent studying literature change if the exogenous amount of coffee increases? Use the IFT to answer this question (you'll either need to use Cramer's rule or to inverse a 3 by 3 matrix). Can you solve for the explicit form of h_L^* as a function of C to verify this answer?

3. How will your maximized GPA change if the exogenous amount of coffee C increases? What if the number of total productive hours T increases? Use Envelop Theorem (the constrained version) to answer this question.

ANSWER:

1. The maximization problem can be written as the following Lagrangian (constrained optimization):

$$\mathcal{L} = \frac{2}{3}\sqrt{C}\sqrt{h_L} + \frac{2}{3}\sqrt{h_E} - \lambda(h_L + h_E - T)$$

We can solve for this maximization problem first. First order conditions:

$$\nabla\mathcal{L} = \begin{bmatrix} 12 - h_L - h_E \\ \frac{1}{3}\sqrt{\frac{C}{h_L}} - \lambda \\ \frac{1}{3}\sqrt{\frac{1}{h_E}} - \lambda \end{bmatrix} = 0 \Rightarrow \begin{bmatrix} h_L^* = \frac{TC}{C+1} \\ h_E^* = \frac{T}{C+1} \\ \lambda = \sqrt{\frac{C+1}{9T}} \end{bmatrix}$$

We can also check if the second-order condition is satisfied at the stationary point. Bordered Hessian:

$$H_L = \begin{bmatrix} \frac{\partial^2 L}{\partial\lambda^2} & \frac{\partial^2 L}{\partial\lambda\partial h_L} & \frac{\partial^2 L}{\partial\lambda\partial h_E} \\ \frac{\partial^2 L}{\partial\lambda\partial h_L} & \frac{\partial^2}{\partial h_L^2} & \frac{\partial^2 L}{\partial h_L\partial h_E} \\ \frac{\partial^2 L}{\partial\lambda\partial h_E} & \frac{\partial^2 L}{\partial h_L\partial h_E} & \frac{\partial^2 L}{\partial h_E^2} \end{bmatrix} = \begin{bmatrix} 0 & -1 & -1 \\ -1 & -\frac{1}{6}\frac{\sqrt{C}}{h_L^{3/2}} & 0 \\ -1 & 0 & -\frac{1}{6}\frac{1}{h_E^{3/2}} \end{bmatrix}.$$

$$det(H_L) = 1 \times \left| \begin{bmatrix} -1 & 0 \\ -1 & -\frac{1}{6}\frac{1}{h_E^{3/2}} \end{bmatrix} \right| + (-1) \times \left| \begin{bmatrix} -1 & -\frac{1}{6}\frac{\sqrt{C}}{h_L^{3/2}} \\ -1 & 0 \end{bmatrix} \right|$$

$$= \frac{1}{6}\frac{1}{h_E^{3/2}} + \frac{1}{6}\frac{\sqrt{C}}{h_L^{3/2}}$$

Since $h*_E > 0$ and $h*_L > 0$. Hence, $det(H_L)$ evaluated at the stationary point is positive, hence we have a local maximum.

2. We can simply calculate $\frac{\partial h_L^*}{\partial C}$ from the explicit solution from part 1:

$$\frac{\partial h_L^*}{\partial C} = \frac{T}{(C+1)^2}$$

The determinant of the Jacobian of the gradient (remember, setting the gradient equal to the vector of zeros gives the f.o.c.), will give the denominator of our expression for $\frac{\partial h_L}{\partial C}$. To find the numerator, let's first write out the Hessian again, but for the second column (that column corresponds to the partial $\frac{\partial}{\partial h_L}$ taken on each term in that column), we instead take the partial with respect to C:

$$\begin{bmatrix} \frac{\partial^2 L}{\partial\lambda^2} & \frac{\partial^2 L}{\partial\lambda\partial C} & \frac{\partial^2 L}{\partial\lambda\partial h_E} \\ \frac{\partial^2 L}{\partial h_L\partial\lambda} & \frac{\partial^2 L}{\partial h_L\partial C} & \frac{\partial^2 L}{\partial h_L\partial h_E} \\ \frac{\partial^2 L}{\partial h_E\partial\lambda} & \frac{\partial^2 L}{\partial h_E\partial C} & \frac{\partial^2 L}{\partial h_E^2} \end{bmatrix} = \begin{bmatrix} 0 & 0 & -1 \\ -1 & \frac{1}{6}\frac{1}{\sqrt{Ch_L}} & 0 \\ -1 & 0 & -\frac{1}{6}\frac{1}{h_E^{3/2}} \end{bmatrix}$$

The determinant of this matrix equals to $-\frac{1}{6}\frac{1}{\sqrt{Ch_L}}$, will give the numerator of our expression for $\frac{\partial h_L}{\partial C}$. Finally, as in the univariate case, we need to multiply by -1. Putting these together:

$$\frac{\partial h_L^*}{\partial C} = -\frac{-\frac{1}{6}\frac{1}{\sqrt{Ch_L^*}}}{\frac{1}{6}\frac{1}{(h_E^*)^{3/2}} + \frac{1}{6}\frac{\sqrt{C}}{(h_L^*)^{3/2}}} = \frac{1}{\frac{\sqrt{Ch_L^*}}{(h_E^*)^{3/2}} + \frac{C}{h_L^*}}$$

You can plug in h_L^* and h_E^* and get the same result at the beginning.

3. Apply Envelope Theorem to Lagrangian to calculate the effect of parameter C and T on the maximized objective function (GPA).

$$\frac{\partial GPA(h_L^*(C,T),h_E^*(C,T);C,T)}{\partial C} = \frac{\partial \mathcal{L}(\lambda,h_L,h_E;C,T)}{\partial C}\,|_{h_L^*,h_E^*,\lambda^*} = \frac{2}{3}\cdot\frac{1}{2\sqrt{C}}\cdot\sqrt{h_L^*} = \frac{1}{3}\sqrt{\frac{T}{C+1}}$$

$$\frac{\partial GPA(h_L^*(C,T),h_E^*(C,T);C,T)}{\partial T} = \frac{\partial \mathcal{L}(\lambda,h_L,h_E;C,T)}{\partial T}\,|_{h_L^*,h_E^*,\lambda^*} = \lambda^* = \sqrt{\frac{C+1}{9T}}$$